

- 6** When activated, T lymphocytes can divide rapidly and secrete proteins called cytokines.

These cytokines will activate different types of lymphocyte.

(a) (i) Which is the function of tRNA in cytokine synthesis?

(1)

- ☐ **A** identifies an amino acid and transports it to a ribosome
- ☐ **B** identifies a base and transports it to a ribosome
- ☐ **C** forms a template for DNA polymerase
- ☐ **D** forms a template for RNA polymerase

(ii) Which organelle is involved in chemically modifying cytokine proteins before they are secreted?

(1)

- ☐ **A** Golgi apparatus
- ☐ **B** ribosome
- ☐ **C** smooth endoplasmic reticulum
- ☐ **D** vesicle

(b) Describe the role of DNA polymerase in DNA replication during cell proliferation.

(2)

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(c) Explain how cytokine genes can be expressed in an activated T lymphocyte.

(3)

(d) T lymphocytes produce different types of cytokine.

Some of these cytokines activate T killer cells and some activate B cells.

Explain why different cytokines can activate different cells.

(3)

- (e) CD45 is a protein found on the surface of lymphocytes. Different forms of the CD45 protein are found on the surface of different lymphocytes.

Describe how different forms of the CD45 protein can be produced from a single CD45 gene.

(3)

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(Total for Question 6 = 13 marks)

